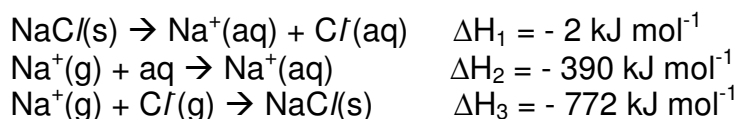


Answer any **four** questions.

1 (a) Given the following reactions

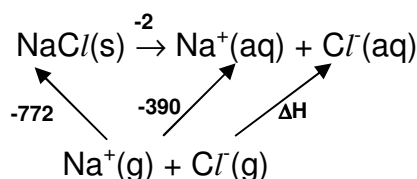


(i) Name ΔH_1 and ΔH_3 .

ΔH_1 = enthalpy change of solution

ΔH_3 = lattice energy

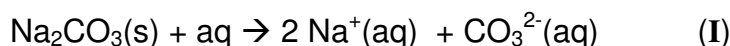
(ii) Draw an appropriate energy cycle to calculate the enthalpy change of hydration of the chloride ion.



$$\Delta H = -772 + (-2) - (-390) = -384 \text{ kJ mol}^{-1}$$

[4]

(b) When 2.12 g (0.02 mol) of sodium carbonate was added to 30.0 cm³ of water, the temperature fell by 5.2 °C.



(i) Calculate the enthalpy change of this reaction per mol of Na₂CO₃. Assume that the specific heat capacities of all solutions are 4.2 J g⁻¹ K⁻¹ and that all solutions have a density of 1.0 g cm⁻³.

$$q = mc\Delta T = \frac{30 \times 4.2 \times 5.2}{1000} = \mathbf{0.655 \text{ kJ}}$$

$$\Delta H = + \frac{0.665}{0.02} = +32.8 \text{ kJ mol}^{-1}$$

(ii) Explain why ΔS of reaction (I) is + 154.3 J K⁻¹ mol⁻¹ and hence calculate the minimum temperature for reaction (I) to be spontaneous.

From reaction, $\Delta n = +1$ or increase in disorderliness, therefore ΔS is positive.

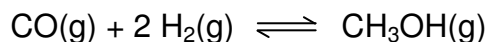
$$\Delta G = 0, \Delta H = T\Delta S$$

$$+32.8 = \frac{T(154.3)}{1000}$$

$$T = 212.6 \text{ K} = 213 \text{ K}$$

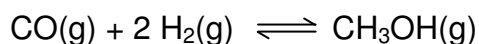
[4]

- (c) CO is used in the manufacture of methanol according to the following reaction:



A 40.8 mol sample of CO was added to 146 mol of H₂. When equilibrium is reached at 400 °C, the mixture contained 30.2 mol methanol and the total pressure was 7.50 MPa.

- (i) Calculate the equilibrium constant K_p for this reaction.



I	40.8	146	0
C	-30.2	-60.4	+30.2
E	10.6	85.6	30.2

$$P_{\text{CH}_3\text{OH}} = \frac{30.2}{126.4} \times 7.5 = 1.792 \text{ MPa}$$

$$P_{\text{H}_2} = \frac{85.6}{126.4} \times 7.5 = 5.044 \text{ MPa}$$

$$P_{\text{CO}} = \frac{10.6}{126.4} \times 7.5 = 0.629 \text{ MPa}$$

$$K_p = \frac{1.792}{(5.044)^2 (0.6290)} = 0.112 \text{ MPa}^{-2}$$

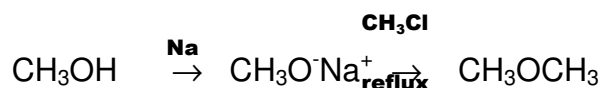
- (ii) In order to increase the yield of methanol, H₂ is added to the equilibrium system. Predict, with reasons, whether this attempt is valid.

With increase in [H₂], position of equilibrium shifts right to decrease the amount of H₂, hence percentage yield of methanol should increase. Therefore attempt is valid.

[3]

- (d) CH₃OH has been found to be useful in the synthesis of other organic compounds. In one process, CH₃OH is used as a raw material for the preparation of organic solvent dimethyl ether, CH₃OCH₃.

Suggest a suitable synthetic route leading to the formation of CH₃OCH₃.

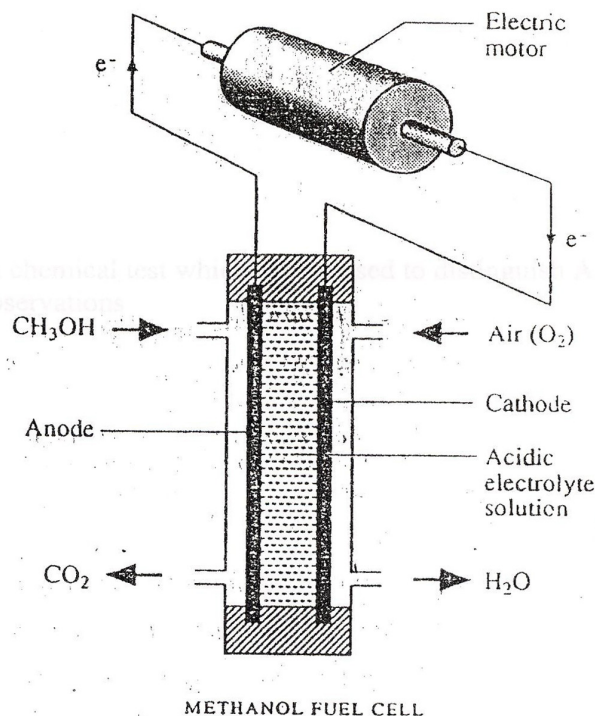


[3]

Answers to YJC 2008 H2 Chemistry Prelim Paper 3

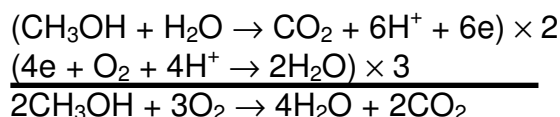
- (e) Scientists are hoping to use methanol to power a fuel cell in the car instead of burning petrol in an internal combustion engine. The fuel cell generates an electric current which drives an electric motor.

One possible design of a methanol fuel cell under investigation is shown below.



The fuel and oxygen are fed continuously to the two electrodes. The electrodes are made of platinum dispersed onto a porous carbon support.

- (i) Write half equations for the reaction which takes place at the electrodes of the fuel cell and hence an overall equation for the cell reaction.



- (ii) Suggest one reason why methanol is a better choice than petrol as a fuel for motor cars.

CH₃OH burns more cleanly, therefore less pollutants given off.

- (iii) One refinement of fuel cell design is to replace the acidic electrolyte solution with a film of solid hydrogen ion conducting polymer. Explain why this would be an improvement.

No problem of leakages when electrolyte changes to a solid film.

- (iv) One fuel cell in use now is the hydrogen-oxygen fuel cell. Give an advantage of this over conventional batteries and cells.

Greater energy efficiency; clean water forms.

[6]

[Total: 20]

- 2 (a) The boiling points of the elements in the second period are shown in the table below.

Element	Li	Be	B	C	N	O	F	Ne
bp/°C	1330	2477	3930	4830	-196	-183	-188	-246

- (i) Explain the variation in the boiling points in terms of their structure and bonding of elements.

Li and Be – metallic bonding, Be stronger metallic bond due to greater of valence electrons.

B and C – very high boiling point due to giant covalent structure

N, O and F – non-metals, weak VDW forces

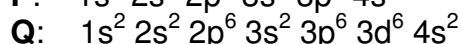
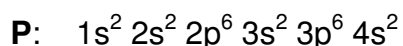
Ne – lowest, weakest VDW being monoatomic

- (ii) Give the equations to show the reactions of Li_2O and CO_2 with water.



[5]

- (b) Consider the two metals whose electron configurations are as follows:



- (i) Compare the atomic radii of the two elements.

Atomic radius of $\text{P} > \text{Q}$

Increase in effective nuclear charge of Q and due to poor screening effect of 3d electrons in Q

- (ii) State and explain in terms of electronic structure, one characteristic property of **Q** (other than that in b(i)) which is markedly different from those of **P**.

Q able to form coloured complexes due to availability of d electrons in partially filled d-orbitals for d-d electron transition. P is unable due to lack of d electrons.

[3]

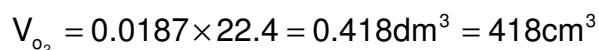
- SiO_2 exists as giant molecular structure with strong covalent bonds between atoms

(d) In an attempt to extract Li from its ore, Li_2O , a student suggested dissolving Li_2O in water, followed by electrolysis of the aqueous solution he had prepared.

- $$\text{H}_2\text{O} + 2\text{e} \rightleftharpoons \text{H}_2 + 2\text{OH}^- \quad -0.83\text{V}$$

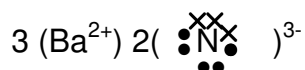
Reduction potential of Li^+ is more positive than H_2O , therefore $\text{H}_2(\text{g})$ forms instead of Li .

- (ii) At the anode, a gas evolved was measured at s.t.p. Calculate the volume of this gas obtained if a current of 2 A was passed through the solution for a duration of 1 hour.

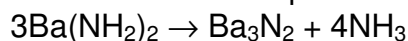


(e) Solid barium amide, $\text{Ba}(\text{NH}_2)_2$ decomposes when heated to give barium nitride Ba_3N_2 and $\text{NH}_3(\text{g})$.

- (i) Draw the dot cross diagram for Ba_3N_2 .



- (ii) Write a balanced equation for the decomposition of barium amide.



- (iii) Based on your knowledge of the thermal decomposition of group II nitrates and by reference to the equation in e(ii), explain why the decomposition is favourable.

Lattice energy of Ba_3N_2 is more exothermic as ionic radius of $\text{N}^{3-} < \text{NH}_2^-$ as well as the magnitude of negative charge in N^{3-} is higher and since

$$\text{L.E.} \propto \left| \frac{q^{n+} q^{n-}}{r^+ + r^-} \right|$$

Ba_3N_2 is energetically more stable.

- (iv) What do you think will be the decomposition temperature of LiNH_2 when compared to $\text{Mg}(\text{NH}_2)_2$? Explain your answer.

ionic radius $\text{Li}^+ < \text{Mg}^{2+}$

charge of Li^+ lower than Mg^{2+}

results in charge density of $\text{Li}^+ \approx \text{Mg}^{2+}$

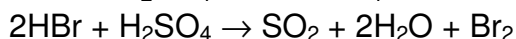
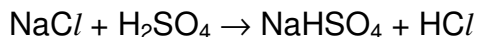
Polarising power of both Li^+ and Mg^{2+} almost the same / Distortion of electron cloud of NH_2^- by both cations almost same

Therefore decomposition temperature of both should be the almost similar.

[6]

[Total: 20]

- 3 (a) Describe what happens when concentrated sulphuric acid is added to solid sodium salts of chloride, bromide and iodide respectively. Give relevant equations.



Observations

For Cl^- , only white fumes of HCl observed. For Br^- , white fumes of HBr first observed followed by brown fumes of Br_2 observed later. For I^- , white fumes of HI first observed followed by violet fumes of I_2 observed later.

[3]

- (b) Under suitable conditions, iodine and chlorine react to give an interhalogen compound, ICl_x . When a pure sample of ICl_x was dissolved in an excess of aqueous potassium iodide, the iodine liberated required 17.1 cm^3 of 0.1 mol dm^{-3} sodium thiosulphate solution for titration whereas the chloride ion liberated required 12.8 cm^3 of 0.1 mol dm^{-3} aqueous silver nitrate for complete precipitation.

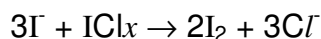
Determine the value of x . Hence, write a balanced equation for the reaction between ICl_x and I^- .

Answers to YJC 2008 H2 Chemistry Prelim Paper 3

$$n_{I_2} = \frac{17.1 \times 0.1}{1000} \times \frac{1}{2} = 8.55 \times 10^{-4} \text{ mol}$$

$$n_{Cl^-} = \frac{12.8 \times 0.1}{1000} = 1.28 \times 10^{-3} \text{ mol}$$

$$\begin{aligned} n_{I_2} : n_{Cl^-} \\ 8.55 \times 10^{-4} : 1.28 \times 10^{-3} \\ 1 : 1.5 \\ 2 : 3 \end{aligned}$$



Therefore $x = 3$.

[4]

- (c) Silver chloride has been found to be a sparingly soluble salt. Given the solubility product of silver chloride is $1.6 \times 10^{-5} \text{ mol}^2 \text{ dm}^{-6}$ at a given temperature.

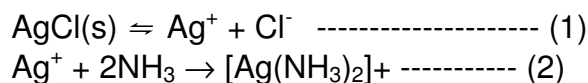
- (i) Calculate the solubility of silver chloride in mol dm^{-3} in water at this temperature.

$$\begin{aligned} [Ag^+][Cl^-] &= 1.60 \times 10^{-5} \\ [Ag^+] &= 4.00 \times 10^{-3} \text{ mol dm}^{-3} \end{aligned}$$

- (ii) Calculate the minimum concentration of dilute HCl(aq) needed to precipitate out silver chloride if 2 cm^3 of the acid is to be added to 10 cm^3 of a solution containing 0.1 mol dm^{-3} of silver nitrate.

$$\begin{aligned} [Ag^+] &= 0.1 \times \frac{10}{12} = 0.0833 \text{ mol dm}^{-3} \\ [Cl^-] &= s \times \frac{2}{12} = 0.1667s \\ (0.1667s)(0.0833) &= 1.6 \times 10^{-5} \\ s &= \frac{1.6 \times 10^{-5}}{(0.1667)(0.0833)} = 1.15 \times 10^{-3} \text{ mol dm}^{-3} \end{aligned}$$

- (iii) When concentrated hydrochloric acid is gradually added to a solution of silver nitrate, a white precipitate is first formed but this dissolves when aqueous ammonia is added in excess. Explain this observation.



$[Ag^+]$ decreases, by LCP, position of equilibrium in eqn 1 shifts right, ppt dissolves.

[5]

Answers to YJC 2008 H2 Chemistry Prelim Paper 3

(d) Given the following third ionisation energies for the elements Ca, Mn and Fe.

Element	Third IE / kJ mol ⁻¹
Ca	4940
Mn	3250
Fe	2960

Comment on the differences in the third ionisation energies for the following pairs.

(i) Ca and Mn

Ca: [Ar]4s²

3rd electron removed from inner complete shell disrupting stable octet configuration, therefore IE is high.

Mn: [Ar]3d⁵4s²

3d and 4s electrons have similar energy hence it is possible to remove 3rd electron and no disruption of stable octet configuration.

(ii) Mn and Fe

Fe: [Ar]3d⁶4s² Mn:[Ar]3d⁵4s²

For Fe, 3rd electron removed involves a paired electron in same orbital experiencing inter-electronic repulsion. Therefore, 3rd IE of Fe < Mn.

[3]

(e)

Ligand	NH ₃	EDTA ⁴⁻
log K (K = stability constant)	2.7	18.6

A solution contains 0.100 mol of aqueous NH₃ and 0.100 mol of EDTA⁴⁻ dissolved in water. 0.01 mol of Ni²⁺ was then added.

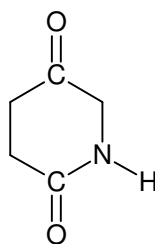
What would you expect to be the main complex ion in the solution? Give a reason for your answer.

Main containing Ni ion solution is complex with edta. The stability constant for complex with edta is greater hence, this complex formed is more stable and energetically favoured. [2]

(f) Give a reason why copper is a transition element.

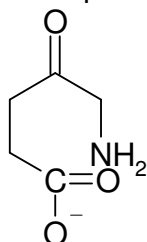
Cu has partially filled d-orbitals, therefore it is a transition element. [1]

(g)

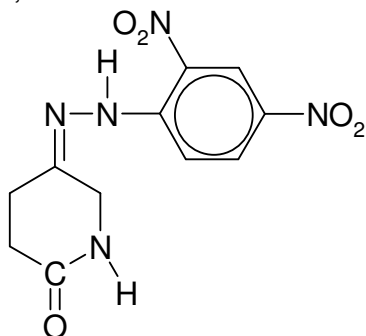


Give the structure of the organic product formed when the above compound reacts with

(i) Hot aqueous NaOH



(ii) 2,4 - DNPH



[2]

[Total: 20]

4 (a) The lattice energies of the potassium halides are given in the table below.

Compound	Lattice energy / kJ mol^{-1}
KF	-801
KBr	-670
KI	-629

Suggest an explanation for the trend shown in the lattice energies.

Ionic radius $\text{F}^- < \text{Br}^- < \text{I}^-$ and $\text{LE} \propto \left| \frac{q^{n+} \cdot q^{n-}}{r^+ + r^-} \right|$

Attraction between K^+ and F^- is strongest while that of K^+ and I^- is the weakest.

[2]

Answers to YJC 2008 H2 Chemistry Prelim Paper 3

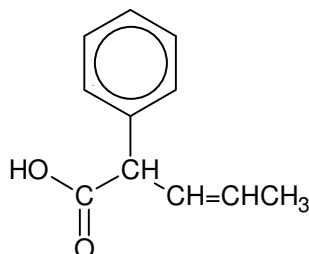
- (b) Nitrogen reacts with chlorine to form NCl_3 but phosphorus can react with chlorine to form both PCl_3 and PCl_5 . Explain the difference in their behaviour.

N: period 2, no available 2d orbital for expansion of octet.

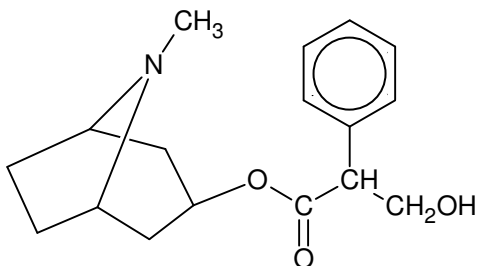
P: period 3, available 3d orbitals allow for expansion of octet.

[2]

- (c) A compound **Y**



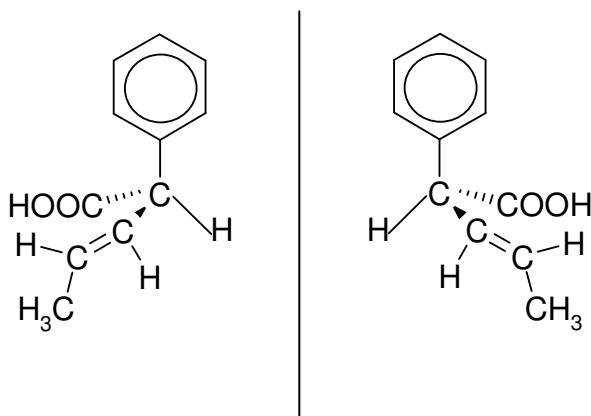
can be derived from atropine

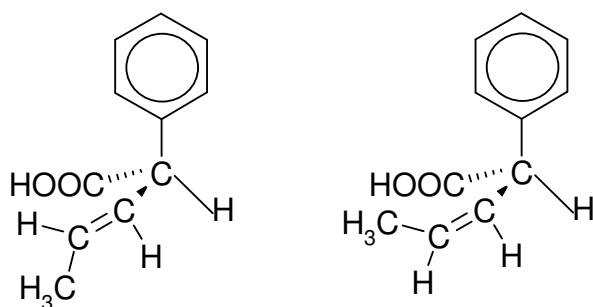


Atropine is a very toxic drug but dilute solutions of it is used during eye examination to dilate the pupil of the eye.

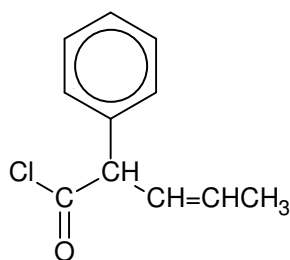
- (i) State the types of stereoisomerism **Y** displays. Draw diagrams to represent them.

Optical isomerism



Geometric Isomerism

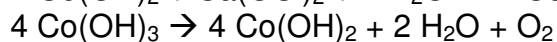
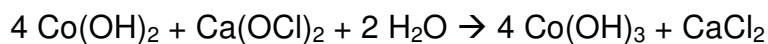
- (ii) Describe reagents and conditions used to produce the following compound from **Y**.



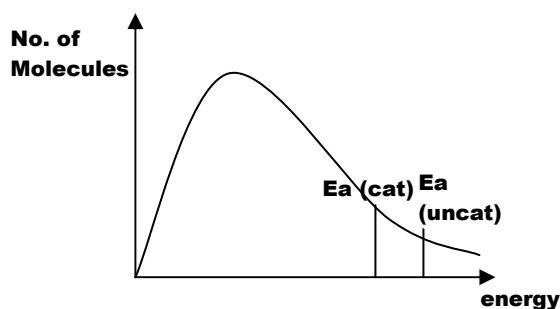
PCl_5 , room temperature

[6]

- (d) The decomposition of bleaching powder (essentially calcium chlorate (I)) occurs by the following steps:



- (i) Using **one** appropriate **diagram**, explain how Co(OH)_2 serves a specific role in the reaction shown.

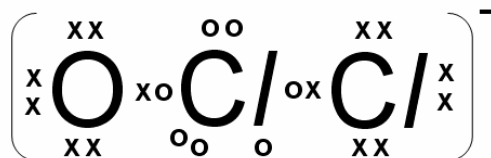


Stable oxidation states of Co^{2+} and Co^{3+} allows it to act as catalyst. Alternative route with lower activation energy as shown by area with activation energy is greater with catalyst. Hence no. of effective collisions increases, rate increases.

[5]

Answers to YJC 2008 H2 Chemistry Prelim Paper 3

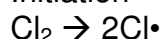
The dot-cross diagram of an anion $\text{OC}/_2^-$ derived from $\text{OC}/^-$ is shown below:



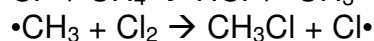
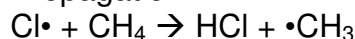
This anion is found to react with methane spontaneously. A sample of methane is reacted with Cl_2 , contaminated with $\text{OC}/_2^-$, in the presence of light.

- (ii) With the aid of equations, give the mechanism in the reaction of Cl_2 with methane.

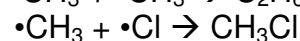
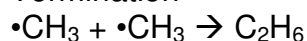
Initiation



Propagation



Termination



[3]

- (iii) In reaction (ii), a small quantity of $\text{CH}_3\text{Cl}_2\text{O}^-$ is formed. Account for this observation with an appropriate equation.



Collision of the above two radicals, $\cdot\text{CH}_3$ and $\cdot\text{Cl}_2\text{O}^-$ give rise to $\text{CH}_3\text{Cl}_2\text{O}^-$ during termination.

[2]

[Total: 20]

- 5 (a) (i) What is meant by the primary structure of a protein?
Unique sequence of amino acids in a polypeptide chain.
- (ii) The folding of the α – helix structure in proteins such as enzymes is caused by interactions of the side-chain (R) groups. Some situations however can destroy the R group interactions. Identify the type of R group interaction that is destroyed in each of the following situation.
- (A) Heating
Hydrophobic interaction and VDW forces between non polar groups
- (B) High acidity
Ionic bonds between charged groups

(iii) 1 mole of a penta-peptide gave the following products on complete hydrolysis: 2 moles of glycine (G), 1 mole of alanine (A), 1 mole of serine (S) and 1 mole of leucine (L). On partial hydrolysis, dipeptides alanylglycine (A-G) and leucylglycine (L-G) were obtained. Partial hydrolysis under another condition gives a dipeptide serylleucine (S-L) and the tripeptide glycylserylleucine (G-S-L).

A G
G S L
S L
L G

[5]

- B** on acidification gave **D**, $\text{C}_3\text{H}_4\text{O}_4$, which reacts easily with aqueous sodium carbonate. 1 mole of **D** gave white fumes of HCl and **E**, $\text{C}_3\text{H}_2\text{O}_2\text{Cl}_2$, when treated with 2 moles of PCl_5 .

$$\begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \\ | \quad | \quad | \\ \text{H}_3\text{C}-\text{C}-\text{C}-\text{C}-\text{CH}_3 \\ | \quad | \quad | \\ \text{OH} \quad \text{H} \quad \text{OH} \end{array} \xrightarrow[\Delta]{\text{I}_2 / \text{NaOH}} 2 \text{CHI}_3 + \text{Na}^+ \text{O}^- \text{C}(=\text{O}) \text{CH}(\text{H}) \text{C}(=\text{O}) \text{O}^- \text{Na}^+$$

$$\downarrow \text{H}_2\text{SO}_4(\text{aq})$$

D

9746/03/YJC2008

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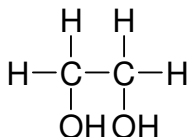
No reaction with CO_3^{2-} , A not a RCOOH .

Hot NaOH , $\text{A} \rightarrow \text{B} + \text{C}$, A is hydrolysed.

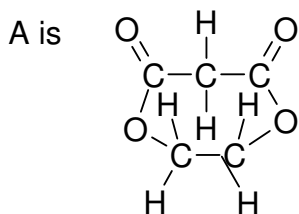
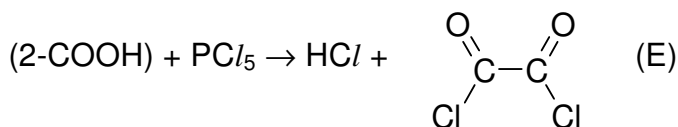
A is an ester.

[O]
C: $\text{C}_2\text{H}_6\text{O}_2 \rightarrow (\text{COOH})_2$

C is a primary alcohol.



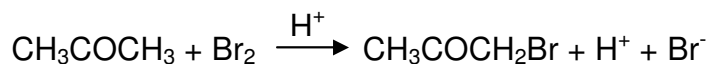
(B) $\text{Na}_2\text{C}_3\text{H}_2\text{O}_4 \xrightarrow{\text{H}^+}$ (D) $\text{C}_3\text{H}_4\text{O}_4$ shows D has 2 $-\text{COOH}$, can react with CO_3^{2-} .



Structures – 5 marks, Any 3 explanations – 2 Marks

[7]

(c) The bromination of acetone is acid-catalysed:



The rate of disappearance of bromine was measured for several different concentrations of acetone, bromine and hydrogen ions at a certain temperature.

Expt No.	$[\text{CH}_3\text{COCH}_3]$ /mol dm ⁻³	$[\text{Br}_2]$ /mol dm ⁻³	$[\text{H}^+]$ /mol dm ⁻³	Rate / mol dm ⁻³ s ⁻¹
1	0.30	0.05	0.05	5.7×10^{-5}
2	0.30	0.10	0.05	5.7×10^{-5}
3	0.30	0.05	0.10	1.2×10^{-4}
4	0.40	0.05	0.20	3.1×10^{-4}
5	0.40	0.05	0.05	7.6×10^{-5}

(i) Determine the order of reaction with respect to each reactant and hence the rate equation.

Answers to YJC 2008 H2 Chemistry Prelim Paper 3

$$\left(\frac{0.3}{0.4}\right)^x = \frac{5.7 \times 10^{-5}}{7.6 \times 10^{-5}} \Rightarrow x = 1$$

$$\left(\frac{0.05}{0.1}\right)^y = 1 \Rightarrow y = 0$$

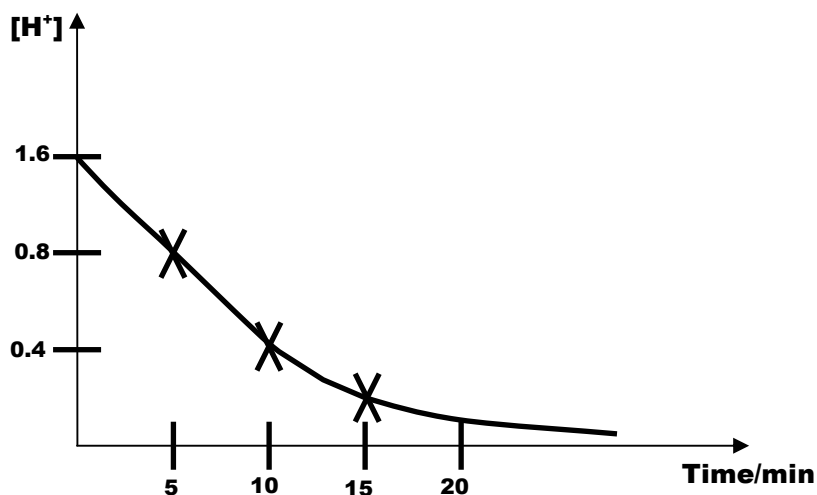
$$\left(\frac{0.2}{0.05}\right)^z = \frac{3.1 \times 10^{-4}}{7.6 \times 10^{-5}} \Rightarrow z = 1$$

$$\text{Rate} = k [\text{CH}_3\text{COCH}_3]^1 [\text{H}^+]^1$$

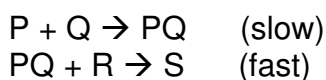
- (ii) In another experiment, the concentration of CH_3COCH_3 used was in excess. As a result, a new rate law was established as follows:

$$\text{Rate} = k' [\text{H}^+]$$

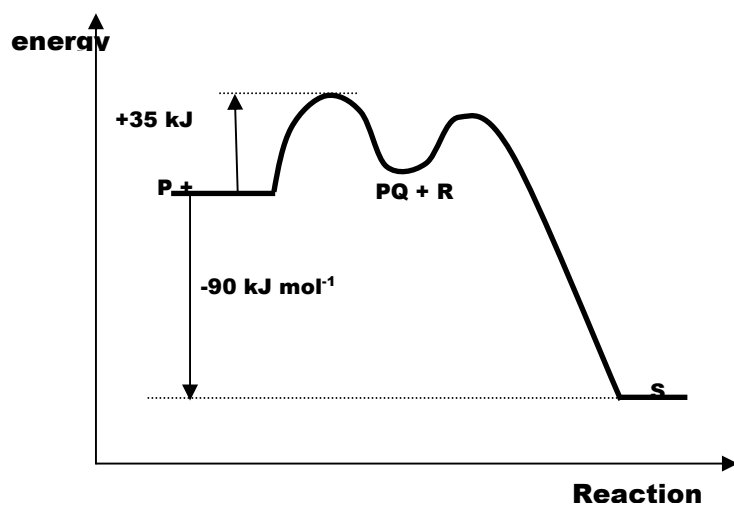
If the initial concentration of H^+ ions is 1.6 mol dm^{-3} and the half-life for the reaction is 5 minutes, sketch a graph of concentration of H^+ against time for the first 20 minutes of the reaction



- (iii) Assigning H^+ as **P** and CH_3COCH_3 as **Q**, the following elementary steps in the mechanism are as shown:



Given that the activation energy of the reaction is $+35 \text{ kJ mol}^{-1}$ and ΔH is -90 kJ mol^{-1} , sketch the potential energy profile diagram for the above reaction.



[8]

[Total: 20]